

Feature-driven characterization of motor speech disorders in nonfluent variant primary progressive aphasia

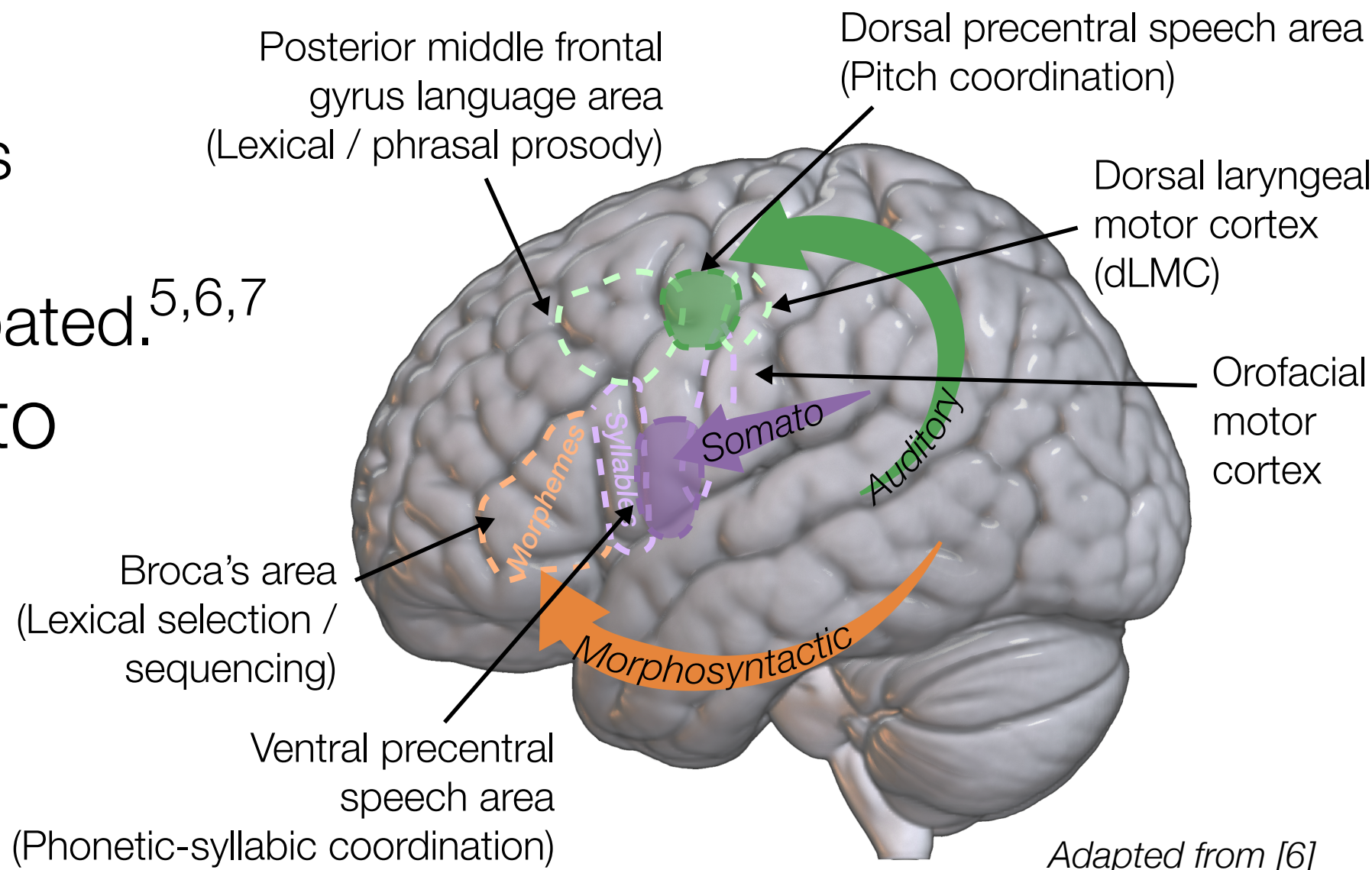
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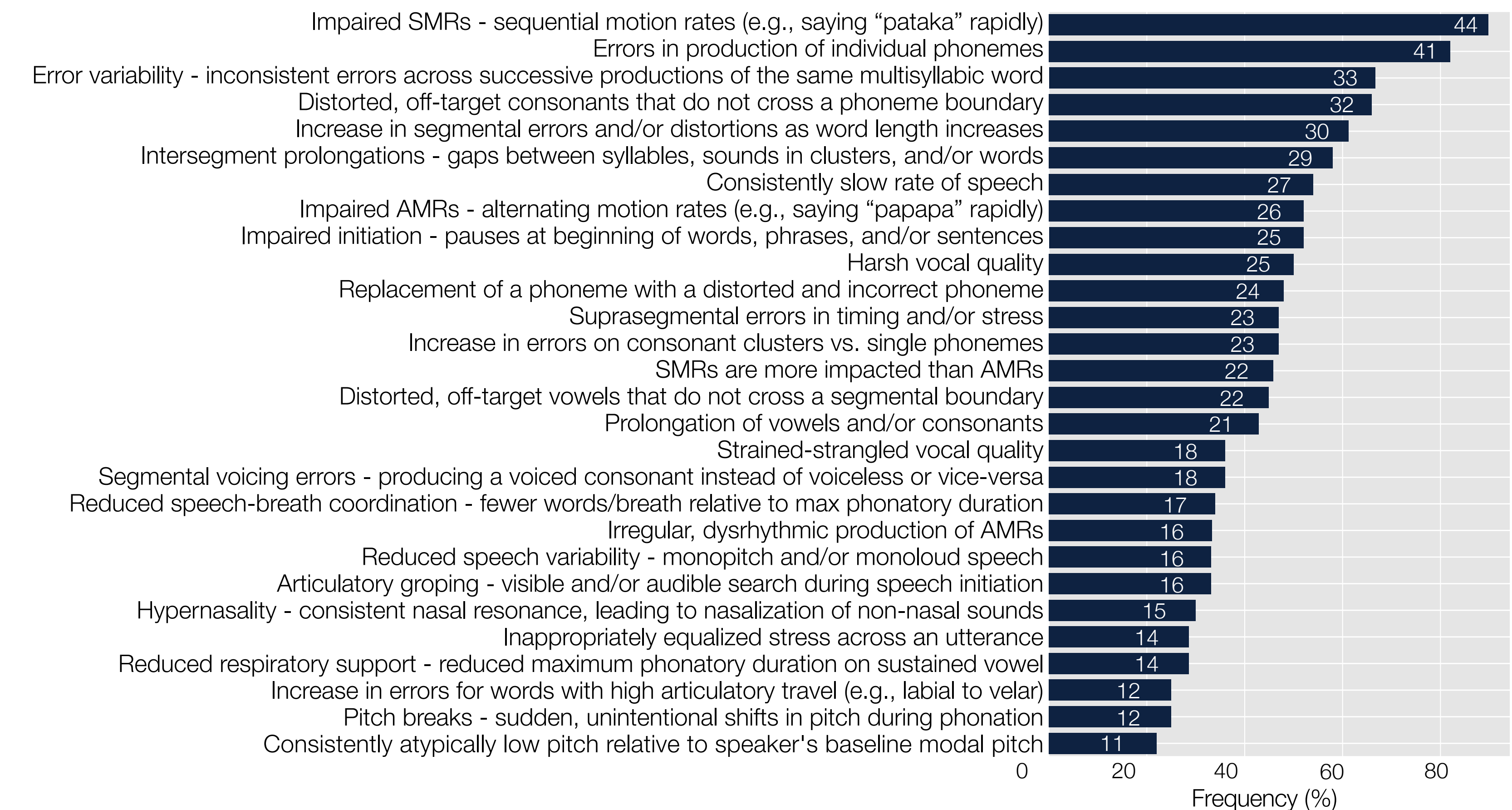
Background

- Nonfluent variant primary progressive aphasia (nfvPPA) is a neurodegenerative syndrome that often involves impaired motor speech.¹
 - Motor speech impairments in nfvPPA-spectrum disorders are quite diverse and the phenotypic boundaries are unclear.^{2,3}
- Many patients present with a mixture of dysarthria and apraxia of speech.⁴
 - Neural and cognitive mechanisms of these symptoms, particularly those related to prosody, are debated.^{5,6,7}
- The goal of this project was to **generate data-driven hypotheses about motor speech profiles in nfvPPA.**



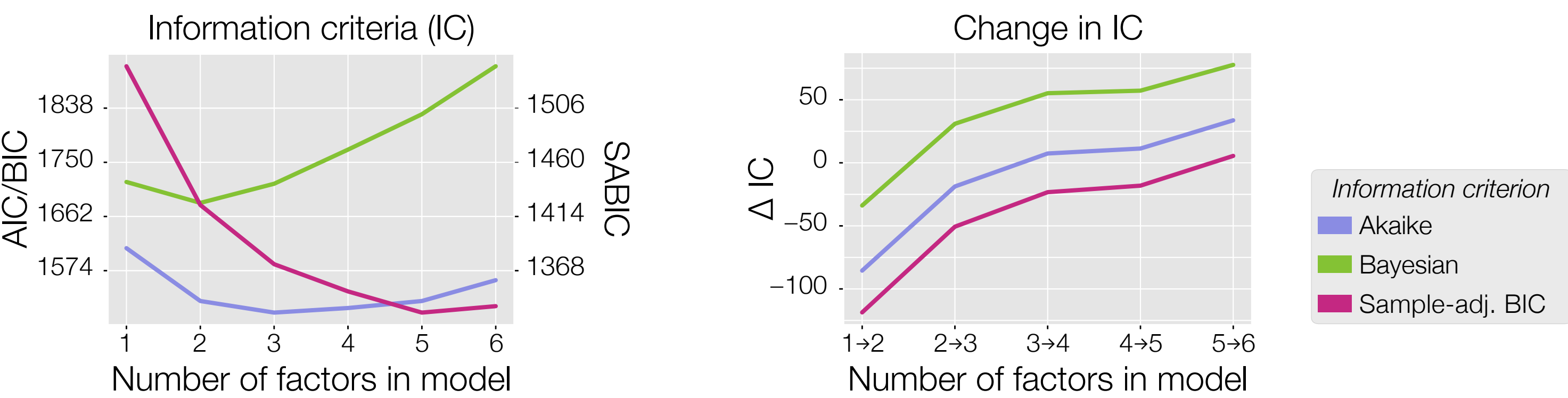
Rating features of motor speech

- Retrospective video review of all patients seen at the Memory and Aging Center 2001-2026 who had any motor speech impairments (AOS or dysarthria severity rated >0).
 - Currently N=50 cases rated. Final N est. at 150
- Fifty features were scored as absent or present by two speech-language pathologists with prior expertise in motor speech disorders.
 - Features represent a near-comprehensive list of plausible motor speech impairments regardless of etiology.
 - 22 excluded from analysis for low prevalence (*scan the QR code for a full list!*)



Exploratory factor analysis

- Using a multidimensional item response theory framework,⁸ we asked how model fit changes with the number of factors to discover latent structure in the co-occurrence of motor speech impairments independent of historic terminology.
- ICs suggest optimal model fit at 2 and 4-5 features, with diminishing returns for higher-dimensional models.



Two-dimensional factorization

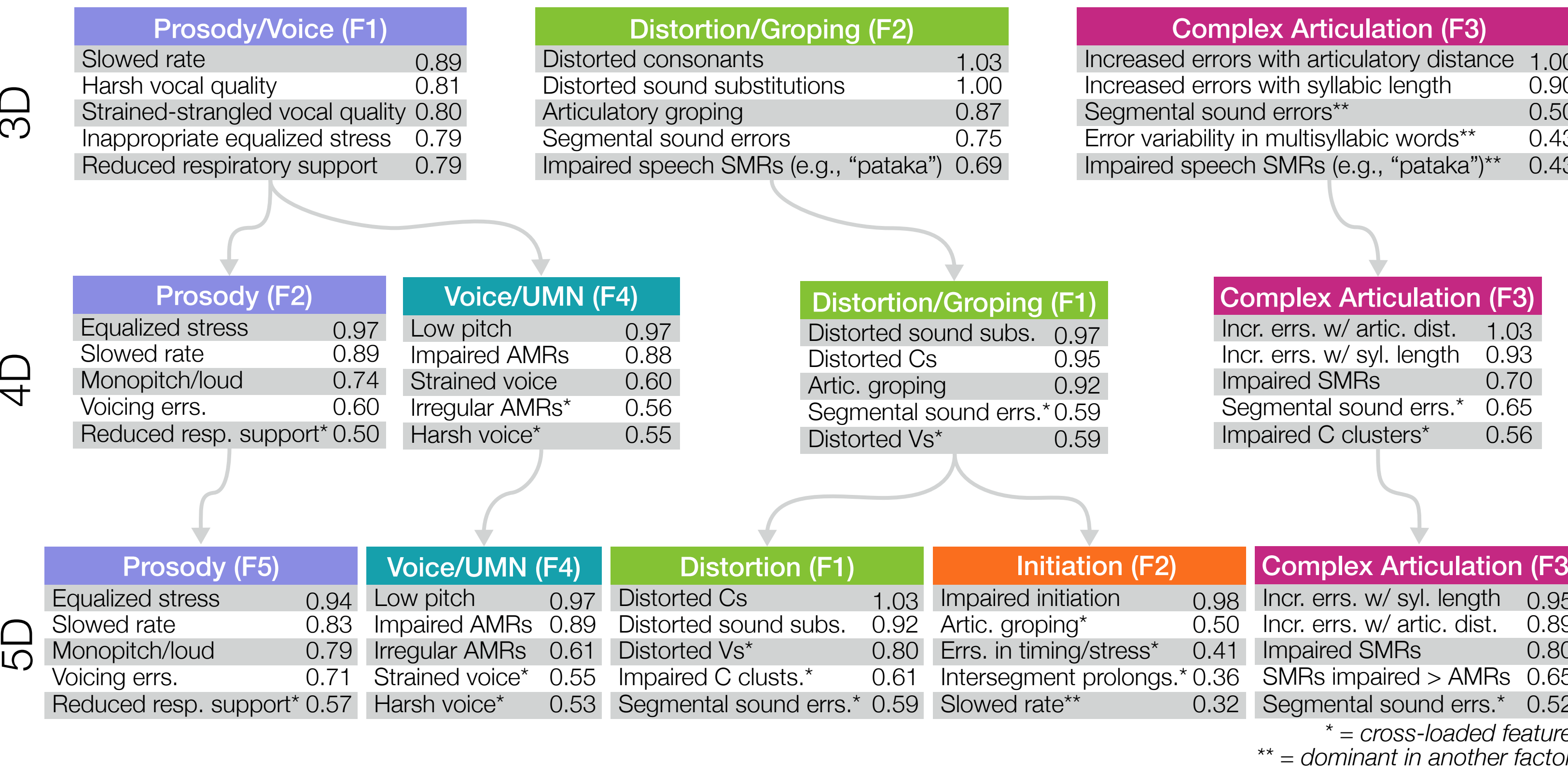
- Feature-wise factor loadings in a 2D solution partially align, but not completely, with a priori motor speech disorders (dysarthria and apraxia of speech).

Strongest F1 loadings	F1	F2
Slowed rate	0.89	-0.09
Strained-strangled vocal quality	0.85	-0.13
Reduced respiratory support	0.81	0.38
Harsh vocal quality	0.79	-0.21
Consistently low pitch	0.73	-0.19
Impaired speech AMRs	0.72	0.02
Equalized stress	0.72	0.17

Strongest F2 loadings	F1	F2
Segmental sound errors	-0.29	1.01
Impaired speech SMRs	0.04	0.90
Distorted sound substitutions	0.18	0.89
Error variability	-0.15	0.86
Distorted consonants	0.23	0.80
Articulatory groping	-0.07	0.76
Intersegment prolongations	0.42	0.74

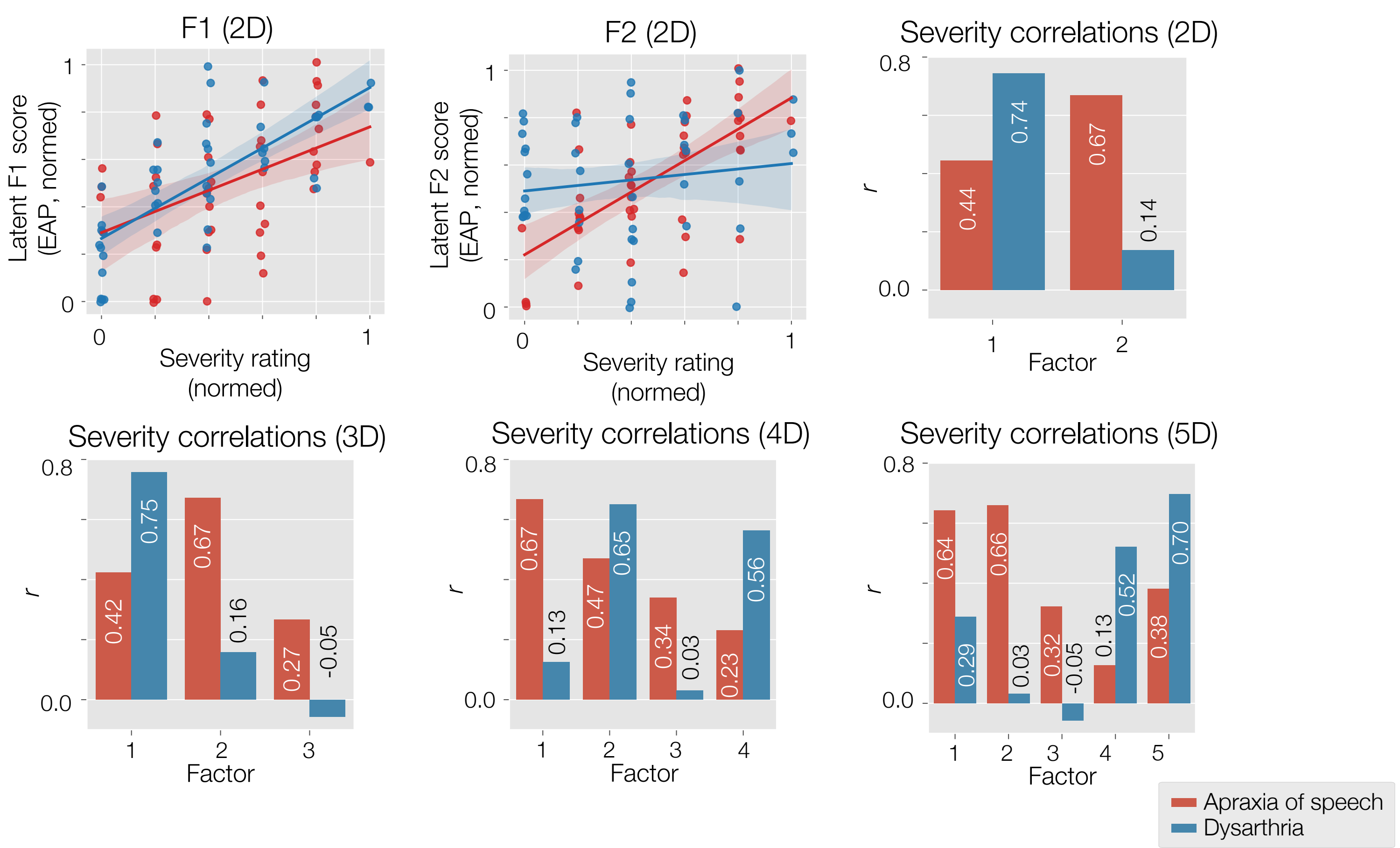
Higher-dimensional factorization

- Progressively finer structure through 3-5 dimensions suggests meaningful subdivisions of motor speech.



Subject-level trait estimation

- In lower-dimensional factorizations, subject-level latent factor scores correlate with independent ratings of AOS and dysarthria severity.
- Higher-dimensional factorizations offer better prediction accuracy yet do not cleanly relate to either severity metric.



Discussion

- Motor speech impairment in PPA is not easily dichotomized into apraxia of speech and dysarthria.
 - Subtyping could improve early diagnosis and pathological prediction.
- Future work aims to map identified factors to patterns of atrophy and underlying pathology.
 - Prosody = dorsal premotor and auditory cortex?
 - Voice/UMN = dorsal motor, basal ganglia, and cerebellum?
 - Distortion = ventral premotor and posterior inferior frontal?
 - Initiation = medial supplementary motor?
 - Complex articulation = anterior insula?

References and Acknowledgements

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